

2022 Republic of China Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part One Student Code



Precautions:

1. This section of the test consists of 8 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part One: **(100 points)** I.

International Competition-Style Questions (50 points in total)

Report 1:

The central nervous system, due to the blood-brain barrier (BBB), is often considered an immune-privileged site, where no immune response occurs. However, the pathogenesis of many neurodegenerative diseases, such as Alzheimer's disease (AD) and multiple sclerosis (MS), is related to inflammation. How do these inflammatory immune mechanisms work?

What is the relationship between immune cells in peripheral blood and inflammation of the central nervous system? Scientists are investigating the origin of immune cells in the meninges. At that time, it was discovered that myeloid cells produced by the skull bone marrow could enter the meninges through channels in the bone, and it was also found that the meninges contained...

Lymphatic vessels (meningeal lymphatics) drain lymph from the brain to the cervical lymph nodes. Antigens in the lymph can be detected by T cells.

Cell identification. Using a mouse model of MS, it was found that bone marrow cells enter the central nervous system in large numbers during brain injury, including those from...

The meninges are more abundant and can regulate immune responses, while bone marrow cells, derived from the blood, are less abundant and are associated with inflammation. In 2021, scientists confirmed...

Cerebrospinal fluid (CSF) can carry antigens from the central nervous system to the dura sinus (DS) venous confluence.

At the flow site, antigen-presenting cells in the dura mater capture antigens and present them to T cells. Single-cell RNA sequencing was used to map the molecular atlases of young and aged dura mater cells, revealing their cell types and genes.

The manifestations were significantly different. Dura mater endothelial cells lacked tight junction proteins and had adhesion molecules on their cell membranes.

(Adhesion molecules) Dura mater stromal cells can secrete the chemokine CXCL12, which attracts T cells with CXCR4 T receptors.

Cells. Animal model studies show that the cumulative activation of immune cells in DS, leading to an immune response that causes neuronal death, may be a contributing factor to MS.

Causes. Most B cells in the meninges originate from the skull bone marrow, where they recognize normal brain proteins, including B cells from the bloodstream.

The number of cells is relatively small but increases with age.

Determine whether the following seven statements (AG) are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the answer sheet.

(Incorrect). A complete wrong answer will earn 0 points. Correct answers to items one through seven will earn 0.5, 1, 2, 4, 7, and 10.5 points respectively.

15 points.

(A) Lymph can flow into the lymph nodes via the meningeal lymphatic vessels, and the antigens brought in the lymph can be recognized by immune cells in the lymph nodes.

Activate the innate immune response in the brain

(B) Single-cell RNA sequencing can identify differences between different cell types, accurately draw evolutionary phylogenetic maps of cell lineages, and...

Prove that a specific sequence has a uniform density distribution on the genome.

(C) Most immune cells in the brain originate from the skull bone marrow and, once inside the brain, can protect it from damage; a small number originate from...

Blood circulation; these immune cells can cause inflammation and damage.

(D) Based on the proteins exhibited by dura mater cells, it is inferred that T cells in the blood can be attracted to the dura mater and attach to it.

On the surface of epithelial cells, adhesion molecules help them cross the intercellular spaces and enter the meninges.

(E) When a virus infects the brain, the permeability of the BBB increases. After infection, the BBB will repair itself, but the immune cells that have already entered will be damaged.

B cells that remain in the brain and produce antibodies against normal brain proteins all enter the brain via the bloodstream.

(F) The brain itself has microglia that can defend against external pathogens and clear away dead cells and metabolic waste from the nervous system.

It is responsible for synaptic pruning and remodeling during neural development; immune cells in the blood do not appear in the brain.

(G) MS T cells in the inflammatory process enter the brain via the bloodstream and attack the myelin sheath as a foreign substance, thus infecting the immune system.

The cytokines and antibodies released by cells can amplify the inflammatory response, leading to BBB breakdown and nerve damage, thus blocking normal nerve function.

Transmission of neural information

Narrative (A)		(B)	(C)	(D)	(AND)	(F)	(G)
Answer							

Report 2:

In February of this year, a cold wave struck, with weather reports describing Taiwan as frozen solid and predicting even lower wind chill. While ambient temperature is an absolute value, can it become a relative value due to somatosensory perception in animals? How do humans perceive temperature or touch? The 2021 Nobel Prize in Physiology or Medicine was jointly awarded to American scholars David Julius and Ardem Patapoutian for their research on the nervous system's perception of temperature and tactile stimuli, respectively. How are external environmental stimuli converted into action potentials or other human-interpretable signals in nerve cells? David Julius, in his research on capsaicin and pain sensation, built a database of millions of DNA fragments. Certain genes correspond to nerve cells that trigger pain, touch, and temperature sensations. One group of genes produces the capsaicin receptor (TRPV1). TRPV1 is an ion channel embedded in the cell membrane; it is activated at temperatures above 43 degrees Celsius, opening the ion channel and converting it into neural signals that trigger pain sensations. Research shows that TRPV1 also responds to proprioception or mechanical force. He then extended his research to temperature receptors, discovering that menthol stimulation of the TRPM8 receptor, a cold-sensitive ion channel, allows it to be perceived. Later, he confirmed that other receptors associated with TRPV1 and TRPM8 respond to different temperatures, explaining how external temperature stimuli are converted into tactile signals in the human body. Ardem Patapoutian, on the other hand, studied how the human body perceives tactile stimuli. Ardem Patapoutian identified a cell line that generates an electrical signal when stimulated by a micropipette. He then screened 72 genes that produce related ion channels and used gene silencing to identify which ones are involved in the transmission of mechanosensitive signals. He discovered the mechanosensitive ion channel (MIC) Piezo1, and further gene comparison revealed Piezo2, which is related to proprioception. Both Piezo1 and Piezo2 are calcium ion channel proteins embedded in the cell membrane with an elastic structure. When the cell membrane is subjected to a certain tension, it straightens from a bent state to a flattened state, opening the calcium ion channel. Ions flow into the cell, converting the mechanosensitive force acting on the cell into an electrical signal that changes cell activity. They play an important role in regulating blood pressure, lung respiration, and bladder contraction in the human body.

Determine whether the following six statements (A-F) are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below. 0 points are awarded for all incorrect answers, and 0.5, 1, 2, 4, 7, and 10 points are awarded for each correct answer (1, 2, 3, 4, 5, 6).

- (A)** Applying sufficient tension to the cell membrane can directly activate Piezo1 and Piezo2 ion channels.
- (B)** By using mice lacking receptor-related genes such as TRPV1 and Piezo1, or through gene silencing, corresponding receptors related to temperature or touch can be identified at the molecular level.
- (C)** TRPV1 is a heat-sensitive receptor activated at high temperatures, while TRPM8 is a cold-sensitive receptor activated at low temperatures. Their mechanisms of action are similar to Piezo1 and... Unlike Piezo2, it does not respond to proprioception or mechanical force.
- (D)** MIC membrane proteins can participate in the transmission of mechanical force signals. The conformational change of the channel after stimulation is the key to the potential changes in peripheral nerves. During the conformational change of the channel, it is not necessarily calcium ions that flow in; potassium ions, sodium ions, and magnesium ions can also produce potential changes.
- (E)** MIC membrane proteins are also commonly found in bacteria, yeast, and plants. In high osmotic pressure environments or under membrane tension, bacteria use Piezo2 to sense mechanical stimuli and open ion channels. The influx of sodium ions converts the mechanical signal into a chemical signal, which then activates other signaling pathways via a second messenger.
- (F)** Somatosensory perception is a general term encompassing touch, pressure, temperature, pain, and proprioception. Different somatosensory modes originate from different types of micro-intensities (MICs) on the cell membrane. Pain sensation in the skin and internal organs, bone growth pain, blood pressure, and general somatosensory perception all have corresponding channels. Studying different channels on the cell membrane can help in the development of different analgesics.

Narrative (A)		(B)	(C)	(D)	(AND)	(F)
Answer						

3.

Receptors and ligands come in many forms, but they all have one thing in common: they pair up tightly.

Receptors recognize only one (or a few) specific ligands, and ligands bind to only one (or a few) target receptors. The binding of a ligand to a receptor alters its shape or activity, enabling it to transmit signals or produce substances directly within the cell.

Changes will occur.

Please determine whether the following seven statements about receptors and ligands (AG) are correct, and fill in the corresponding column in the answer sheet with "T".

(F represents correct) or (F represents incorrect). A score of 0 is awarded for all incorrect answers. Correct answers to items one through seven will receive the following points:

Scored 0.5, 1, 2, 4, 7, 10.5, 15 points.

(A) There are many types of receptors, mainly divided into intracellular receptors, which exist inside the cell (cytoplasm).

(Or in the cell nucleus); cell surface receptors, mainly found in the cell membrane or endometrial system.

(endomembrane system) y

(B) There are many different types of ligands, but they must have a protein (or short peptide) as the main backbone structure.

(backbone structure); or hydrophobic molecules such as sterols; other small molecules due to their interaction with receptors

Insufficient affinity prevents binding to the receptor, leading to signal transmission failure.

(C) The ligands of intracellular receptors are mainly hydrophilic molecules because the cytoplasm is primarily a hydrophilic environment.

(D) Cell surface receptors must bind to extracellular surface ligands; therefore, structurally they are mostly membrane-anchored proteins.

In this type of signal transduction (anchored protein), the ligand needs to cross the cell membrane and enter the cell to induce...

A series of chain reactions

(E) Ligand-gated ion channels: Ion channels that can only be opened after binding to a ligand.

Therefore, the transmembrane-spanning region of this type of cell surface receptor forms a hydrophilic...

Hydrophilic channels allow ions to pass smoothly across the cell membrane.

(F) Enzyme-linked receptors (ELRs) are cell surface receptors with enzymatic functions, specifically receptor tyrosine kinases.

Receptor tyrosine kinase (RTK) is a common enzyme-linked receptor on human cells. The kinase on the RTK...

Transferring phosphate groups to ligands amplifies signal transduction.

(G) Peptide ligands are water-soluble ligands, commonly used in neurotransmitters such as enkephalins.

For example, peptides (enkephalins) can have a molecular length of over a hundred amino acids.

Narrative (A)	(B)	(C)	(D)	(AND)	(F)	(G)
Answer						

4.

Prokaryotic cells (such as bacteria) are organisms with a simple structure; lacking an endomembrane system, they are often described as undivided rooms.

Eukaryotic cells, on the other hand, are neatly separated by layers of membrane-like structures.

Each compartment is kept under independent conditions to perform the specific conditions required for its metabolic work. This is because within a single cell...

The ability of eukaryotic cells to withstand different environments enables them to perform more complex metabolic reactions than prokaryotes, thereby increasing their cell size.

Larger than a prokaryotic cell.

Please determine whether the following six statements (A-F) regarding eukaryotic cells are correct. Fill in the corresponding column in the table below with **T** (for correct) or **F** (for incorrect). 0 points are awarded for all incorrect answers. Correct answers to items 1, 2, 3, 4, 5, and 6 are awarded 0.5, 1, 2, and 4 points respectively.

7. 10 points.

(A) Genetic material forms multiple linear chromosome structures and has a specific number of chromosomes.

(B) Ribosomes are free in the cytoplasm and are mainly responsible for protein synthesis.

(C) The smooth endoplasmic reticulum is primarily responsible for the synthesis of lipids and steroid hormones.

(D) Centrioles are cylinders composed of nine microtubule triplets, responsible for the separation of chromosomes during cell division.

A cell lacking this structure cannot complete cell division.

(E) Plants have their own supporting extracellular structure called the "cell wall," which is a polysaccharide organic molecule composed of glucose units.

Cellulose is a large molecule secreted by cells.

(F) Plant cells are surrounded by cell walls, unlike animal cells which can connect to each other through cell membranes; they can only connect through specialized junctions.

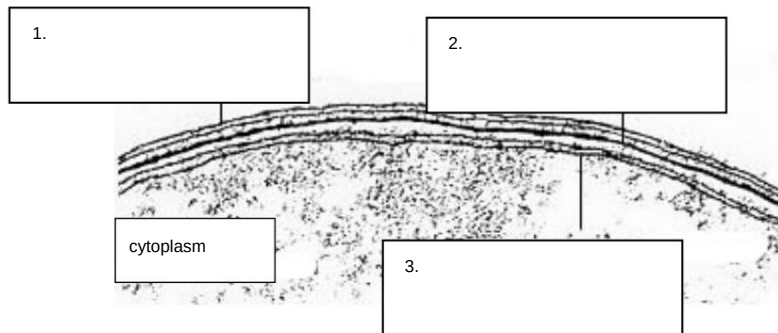
Plasmodesmata (pl.) facilitate the diffusion of small molecules between cells, including proteins.

The complex structure of the particles prevents them from passing through smoothly, thus achieving the goal of material control.

Narrative (A)		(B)	(C)	(D)	(AND)	(F)
Answer						

II. The image below is an electron microscope image of the cell wall of a Gram-negative bacterium. Please fill in the blanks with the name of this structure.

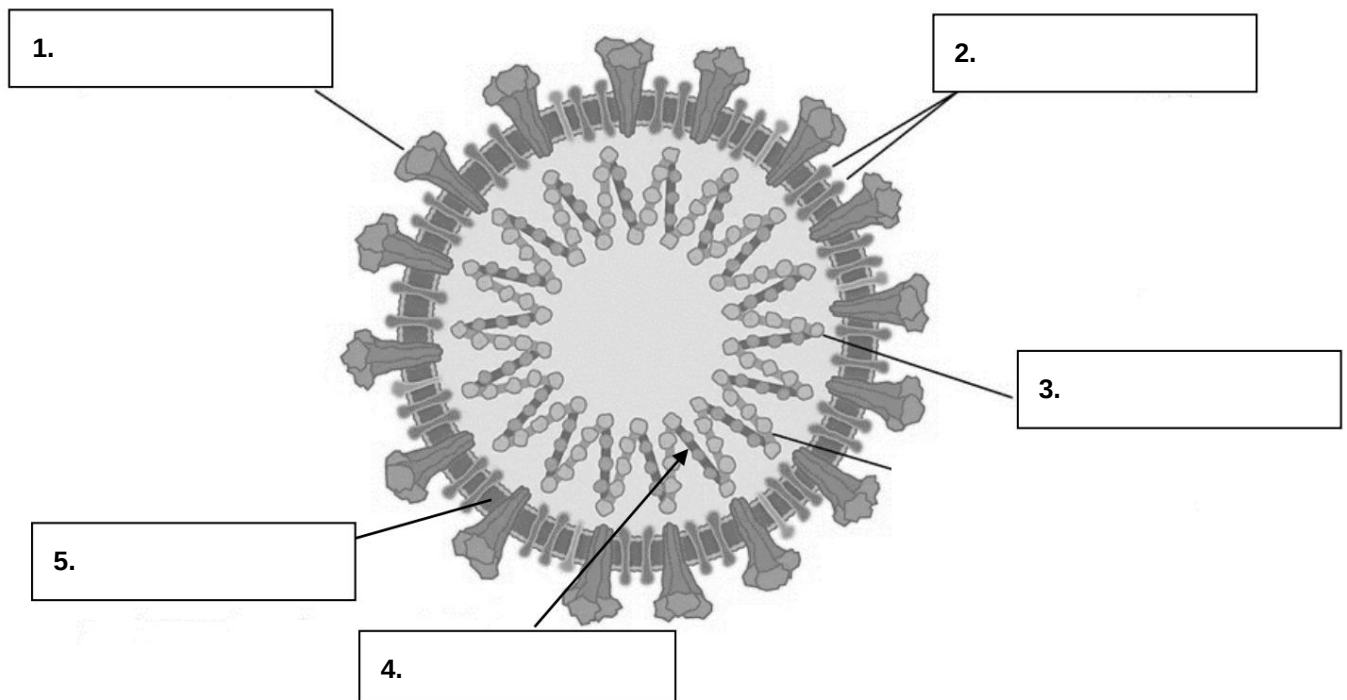
(2 points per blank, 6 points in total)



III. Please compare the structural similarities and differences between ribosomes in prokaryotic and eukaryotic cells (excluding chemical composition). (4 points)

IV. Please compare the similarities and differences in the structure and mode of movement of flagella between prokaryotic and eukaryotic cells. (5 points)

V. The pathogen of COVID -19 is a coronavirus, **SARS-CoV-2** . The following diagram is a model of **SARS-CoV-2** .
Please fill in the names of each structure. (2 points per blank, **10 points in total**)



VI. True or False Questions (The following are descriptions of cell signal transduction. Please answer **O** or **X** according to the question.)

_____ 1. G proteins bind to and are regulated by guanine nucleotides.

_____ 2. When bound to GDP, the G protein is activated.

_____ 3. All G proteins require the activation of G protein-coupled receptors (GPCRs).

Only then can messages be transmitted to the outside world.

_____ 4. Cells in multicellular organisms use extracellular vesicles for signal transduction.

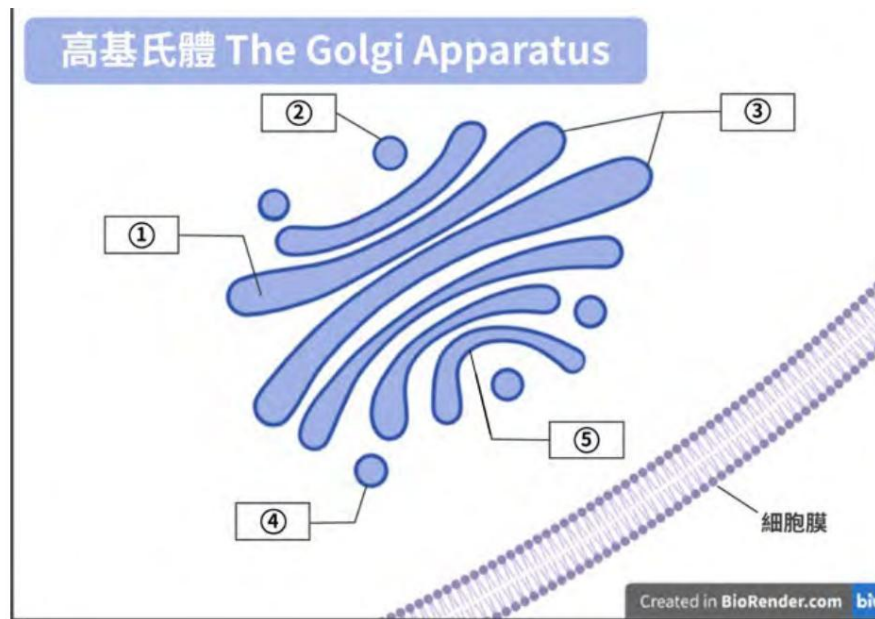
_____ 5. Single-celled organisms can communicate with each other through volatile organic compounds (VOCs).

Distance-based information transmission molecules.

7. Why are second messengers like **cAMP** needed for cellular communication?

(Essay Question)

8. The image below shows Golgi apparatus .



Please fill in the name according to the structural location of the high-millimeter-gravimetric structure.

ÿ _____

ÿ _____

ÿ _____

ÿ _____

ÿ _____

9. What is the function of Gorgistii in plant cells? (Essay Question)

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Theoretical Assessment Test Questions

Part Two Student Code



Precautions:

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Part Two: (100 points)

I. Choose the most suitable answer from the options and fill it in the blank below (2 points each, 12 points in total)

(1) What are the characteristics of the phylum Cyanophyta?

- | | |
|--------------------|---------------------|
| A. Eukaryotic | B. Heterocysts |
| C. Monospore | D. Floridean starch |
| E. Carposporophyte | F. Tetraspores |

(2) What is the structure or characteristic of the alternation of generations in Porphyra?

- | | |
|--------------------|----------------------|
| A. False Branching | B. Spores (Gonidium) |
| C. Carpospores | D. Coenocytic |
| E. Conceptacle | F. Conjugation (3) |

Which is a species or product of the Rhodophyta phylum:

- | | |
|---------------------|---------------------|
| A. Anatoxins | B. <i>Spirulina</i> |
| C. <i>Chlorella</i> | D. Carrageenan |
| E. Microcystins | F. Algin |

(4) Which of the following is a prokaryotic algae?

- | | |
|--------------------------------------|---|
| A. <i>Halimeda</i> (a type of algae) | B. <i>Chlorella</i> |
| C. <i>Polysiphonia</i> | D. <i>Chlamydomonas</i> (a type of single-celled algae) |
| E. <i>Prochlorothrix</i> | F. <i>Volvox</i> |

(5) What are the characteristics of Charophyceae?

- | | |
|---------------------|---------------------|
| A. Statospores | B. Wakame |
| C. Coccoliths | D. Pigmentation |
| E. Spore (Gonidium) | F. Phragmoplast (6) |

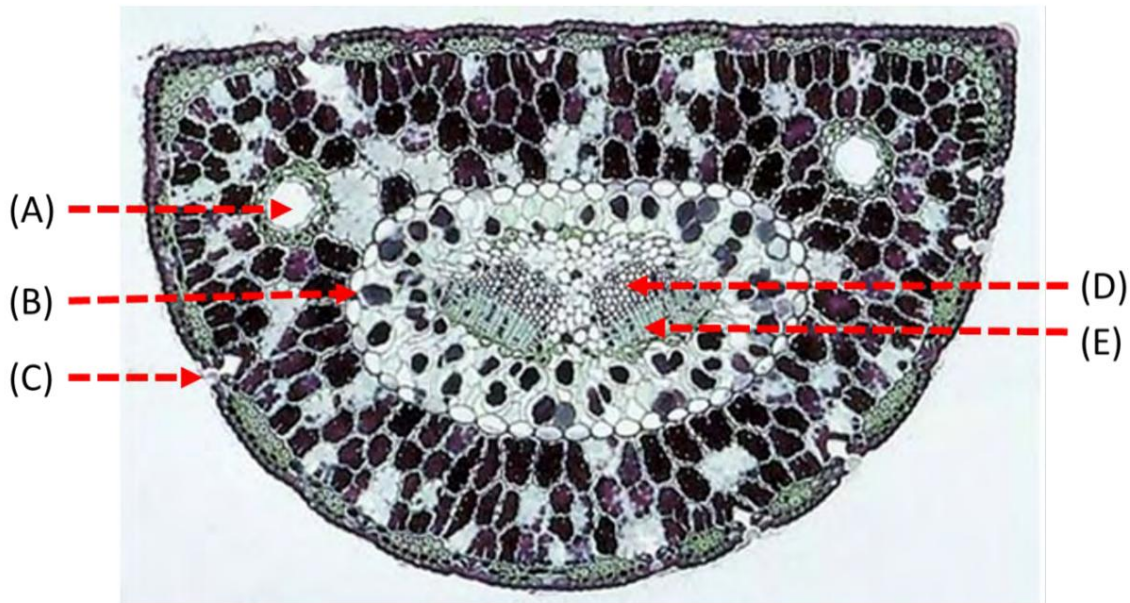
What is a characteristic of the alternation of generations in Ulva?

- | | |
|----------------------------|----------------------|
| A. Isogametes (Isogametes) | B. Spores (Gonidium) |
| C. False Branching | D. Tetraspores |
| E. Conjugation | F. Heteromorphic |

(1)	(2)	(3)	(4)	(5)	(6)

II. Below is a cross-section of a pine needle. Please fill in the structure of the parts marked (A) to (E) (1 point each, 5 points in total) and briefly describe its structure.

Function (2 points each, 10 points in total)



The name of the problem structure	Brief description of functions	
(A)		
(B)		
(C)		
(D)		
(AND)		

III. Flower polymorphism in plants has significant evolutionary implications. The signals provided by the angiosperm corolla are crucial for pollinators to locate food rewards.

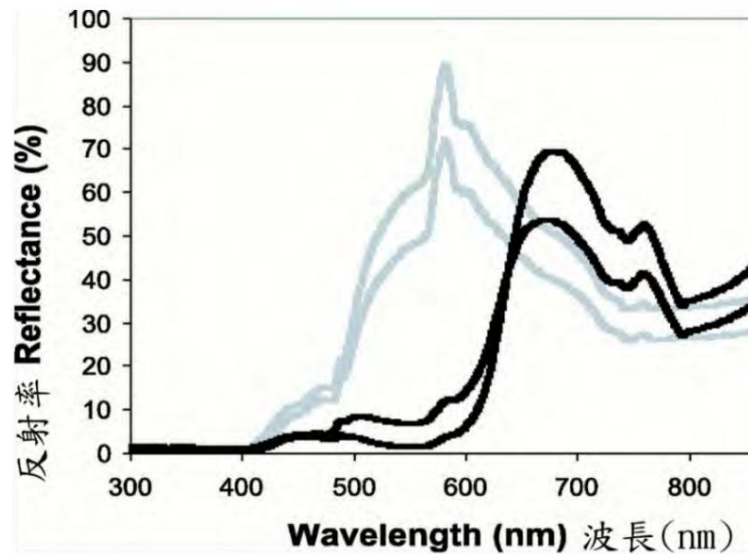
High proportions of flower color polymorphism have been recorded in many orchid species.

These polymorphic species lack nectar and their pollen forms pollinia, thus offering no reward to their pollinators (no *food* incentive). The elderflower (*Dactylorhiza sambucina*) is an example of this type, possessing individuals with both yellow and purple flowers (Fig. 3-1). Neither of these two colored flowers has nectar glands, making them a non-rewarding orchid species for pollinators. Individuals with yellow and purple flowers exist throughout the European population, and the frequency of yellow and purple flowers within the population dynamically varies from year to year, consistently maintaining this flower color polymorphism.

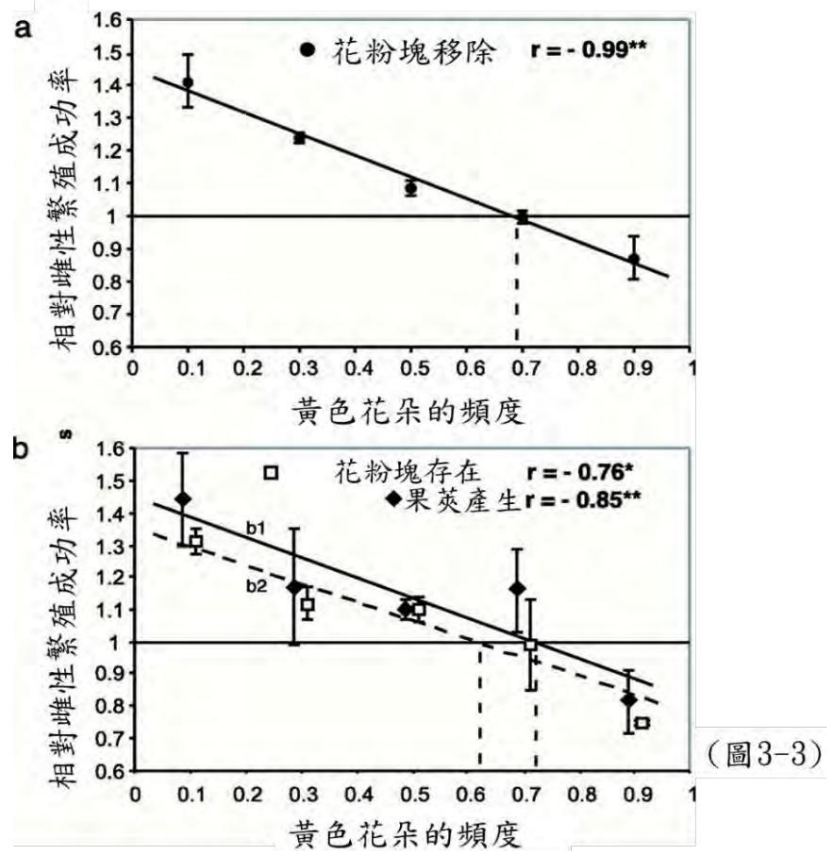


(Figure 3-1) Flowering pattern of a natural population, showing the coexistence of yellow and purple flowers.

To determine the mechanism maintaining flower color polymorphism, researchers examined the reflectance spectra of the two flower colors. The study found that the measured spectral bands of these two colors included the wavelengths of reflectance spectra within the visual sensitivity range of their insect pollinators (Figure 3-2), indicating that insect pollinators are attracted to both flower colors.



(Figure 3-2) The reflectance spectra of the two flower colors are used as wavelength maps. Yellow flowers are light gray, and purple flowers are black. The two spectral lines given for each color (light gray and black) in the figure represent the spectral range obtained from 10 individuals. The researchers constructed 10 simulation analyses in a natural population to investigate the frequency of flower color and the frequency of pod production within the population, in order to test the correlation between flower color frequency and relative reproductive success rate. Through pollen removal and presence (without removal) experiments on yellow flower individuals, the results are shown in (Figure 3-3a) and (Figure 3-3b), respectively.



(Figure 3-3) Flower color and fruit set success rate experiment. The solid line represents the pollen mass (removal/presence) trend line, and the dashed line represents the pod production trend line. Insect pollinators were allowed to randomly visit and pollinate flowers. The X-axis represents the frequency of yellow flowers in the natural population, and the Y-axis represents the relative female reproductive success rate.

• Based on the information above, answer the following questions:

1. Briefly describe the results that can be concluded after the pollen mass is removed, as shown in Figure 3-3a. (3 points) What is the significance of this design in the scientific method?

What? (3 points) How do you determine if it is reasonable? (2 points)

2. Briefly describe the results that can be drawn by comparing the solid and dashed lines in Figure 3-3b, where pollen masses are present. (4 points) How can you determine whether this

comparison is reasonable? (4 points)

3. Based on the above information, including the results in Figures 3-2 and 3-3, it is inferred that the frequency of yellow and purple flowers changes in natural populations.

What are the circumstances surrounding the movement? (3 points) What are the possible conclusions drawn from this research? (4 points)

IV. Exploration of the Floral Organ Identity Gene-ABCDE Model: Plants

Regarding the developmental pattern of reproductive organs, scholars (A. Kanno et al., 2003) previously proposed the ABC model, as shown in Figure A below. Classical... The ABC model effectively utilizes genes of classes A, B, and C to explain the molecular mechanisms of floral organ development and can broadly explain... Variations in plant floral organs caused by mutations in homeotype genes. Subsequently, some studies indicated that D and E type genes seem to be... It also participates in floral organ formation. Extensive research into the characteristics of floral organ genes and proteins has improved the ABC model, leading to the current understanding... The previously widely accepted ABCDE model. According to research by Couto, Márcio & Amorim, Maria (2019), they proposed... The ABCDE model is shown in Figure B below. Please answer the following questions regarding the genes of floral organs based on your/her knowledge of botany.

Answer the following questions based on relevant knowledge of physics and the information in the diagram below (20 points).

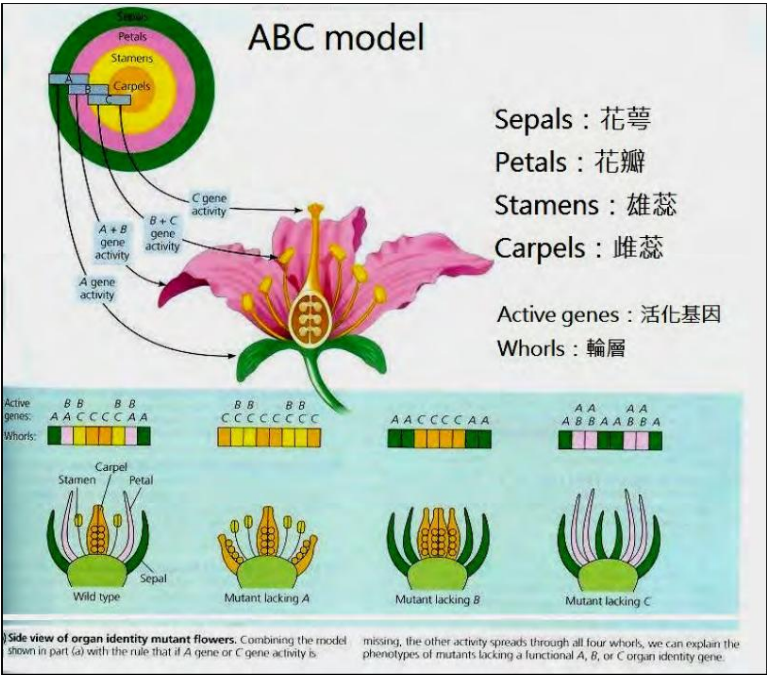


Figure A: ABC model

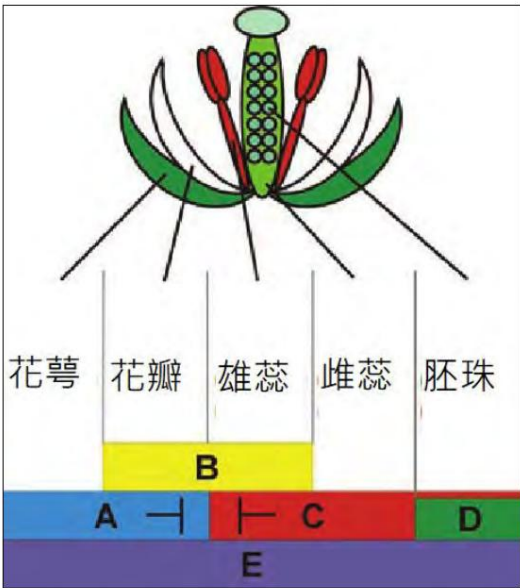


Figure B: ABCDE model

1. Corresponding to the ABC model diagram, the morphological traits of flowers exhibited by the active genes of A+C.

for _____ This corresponds to the ABCDE model diagram, B+C+E

The traits expressed by the active genes are _____ (5 points)

2. Among the genes governing floral organs, the ABC model represents a complete flower.

The active gene is _____ ABCDE model

active genes _____ (5 points)

3. The image below shows the flowers and fruits of the common street tree, the golden shower tree (*Symplocos fragilis*), corresponding to the ABCDE model diagram.

active genes _____ (5 points)



4. Based on the ABCDE model diagram, it is inferred that the active genes containing high levels of auxins are A, B, C, D, and E.

Which gene(s)? And why? (5 points)

V. Investigation of the oxidative pentose phosphate cycle (OPPC): Plant growth and development

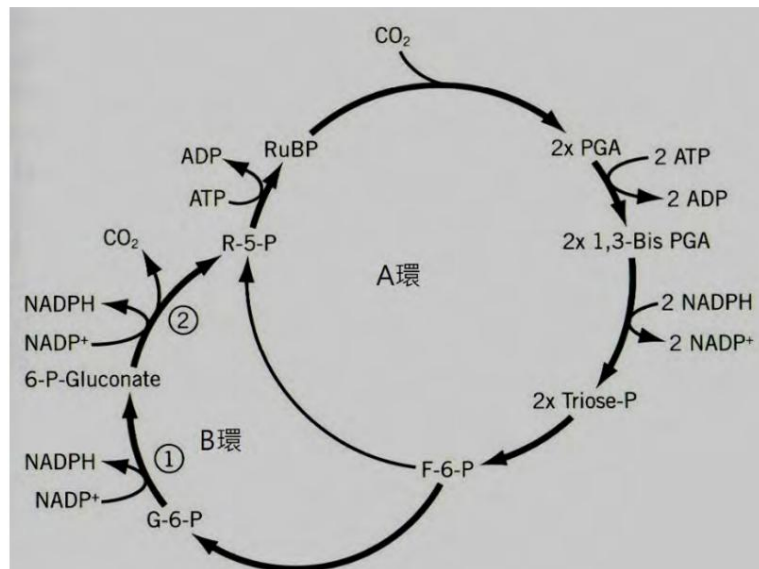
During environmental adaptation, energy, the carbon skeleton, and related metabolic reactions all need to be integrated and coupled, including the oxidation of pentose sugars and phosphorus.

The acid cycle is one example. The diagram below includes ring A and ring B, where ring B represents the OPPC cycle (PGA: phosphate group).

Oleic acid; Triose-P: three-carbon phosphate; F-6-P: fructose-6-phosphate; R-5-P: ribose-5-phosphate; RuBP: 1,5-bisphosphate

Ribulose; G-6-P: glucose-6-phosphate; gluconate (gluconate). Regarding the role of plant OPPC...

Please answer the related questions based on your/your knowledge of plant physiology and the information in the image below (30 points).



1. What are the differences in the production or consumption of ATP and NADPH between rings A and B in the diagram? What are the differences in the carbon dioxide content? (5 points)

2. What type of cycle is ring A in the diagram also called? What is its function? (5 points)

3. What role does NADH, an analogue of NADPH, play in cellular respiration? What roles do they play in biochemical synthesis?

(10 points)

4. What role does ring B play in the diagram? Under what circumstances will ring B show a significant effect? (10 points)

Theoretical Assessment Test Questions for the **2022** Republic of

China Biology Olympiad

National Team Selection Camp

Part Three

Student Code



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9

Part Three: (100 points) I.

Multiple Choice Questions

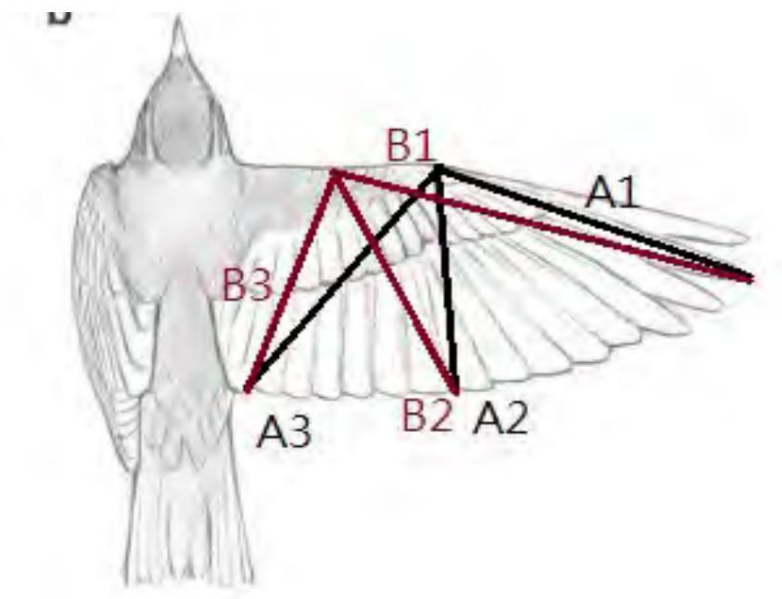
1. The shape of bird eggs has long been a source of curiosity. The eggs of most bird species fall somewhere between three extreme shapes: spherical, oval, and conical. However, humans still know very little about the mechanisms underlying these egg shapes. There are many hypotheses regarding spherical and conical eggs. Choose the most reasonable description from the following options (8 points).

Answer: _____ (Fill in the code)

- (A) Spherical bird eggs have the largest surface area to volume ratio, which can increase the respiration efficiency of the embryo.
- (B) Spherical bird eggs have more uniform physical strength in all directions, which reduces the risk of breakage.
- (C) Spherical bird eggs of the same size require the least amount of eggshell material, thus saving on investment in egg production.
- (D) Spherical bird eggs have the most uniform pore density, which may be beneficial for gas exchange in bird eggs.
- (E) The same number of spherical bird eggs can be arranged most closely, which can increase incubation efficiency.
- (F) Conical bird eggs have the smallest turning radius when rolling, which helps reduce the chance of the eggs rolling off cliffs.
- (G) The higher pore density at the blunt end of the conical egg may be beneficial to the neural development of precocious chicks.
- (H) When multiple cone-shaped bird eggs cluster together, the portion in contact with the base is larger, which may make them more impact-resistant.

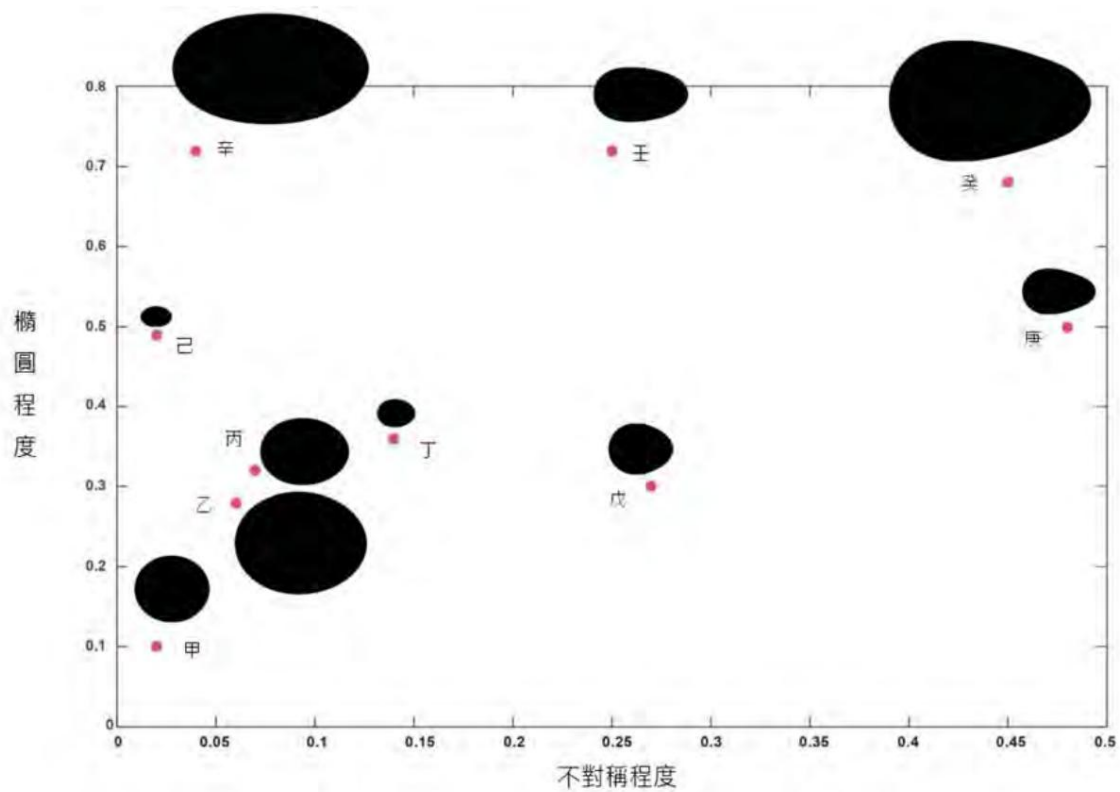
II. Fill in the blanks and answer

questions 1. A 2017 study published in Science measured the morphology of nearly 50,000 bird eggs from over 1,400 bird species. Comparing egg shapes across different bird lineages, and considering factors such as feeding habits, nesting locations, body size, and flight ability, the study successfully used mathematical models to explain the differences in egg shapes and identified the correlation between egg shape and key characteristics of different bird species. The results showed that egg shape is primarily influenced by flight ability (indicated by the wing index (distance from the wrist joint to the tip of the longest primary flight feather / distance from the wrist joint to the tip of the first secondary flight feather)). Birds with better flight ability and those requiring longer distances tend to have eggs that are more conical or oval in shape.



1. According to the diagram above, the two line segments that should be measured for the wing index are:

y _____ / _____ (2 points)



The solid black circles in the image above represent the silhouette of a bird's egg. Note that the relative size of the solid circles can only roughly reflect, but cannot linearly correspond to, the relative size of the original bird's egg.

2. Please select the silhouette of the bird egg that best matches the following bird species (5 points).

(1) Hummingbird: () (2) Domestic

chicken: () (3) Black-crowned egret:

() (4) Golden plover: () (5) Collared

scops owl: () _____

3. Draw the above diagram with $X = 0.25$ and $Y = 0.4$ into four quadrants. Based on the research results, in which quadrant would the eggs of flightless birds such as ostriches, cassowaries, sandpipers, and penguins fall? Why? (2 points)

4. Compared to other flightless birds, penguin eggs are noticeably less asymmetrical in shape. Why?

(3 points)

5. Compared to other flightless birds, kiwi eggs are the most oval. Why?

(5 points)

III. Fill in the blanks and answer questions

pH levels are closely related to individual physiological functions. The normal pH level of human blood is maintained at approximately 7.4.

The lungs and kidneys play a crucial role in controlling blood pH levels. When the kidneys or lungs malfunction, it can lead to...

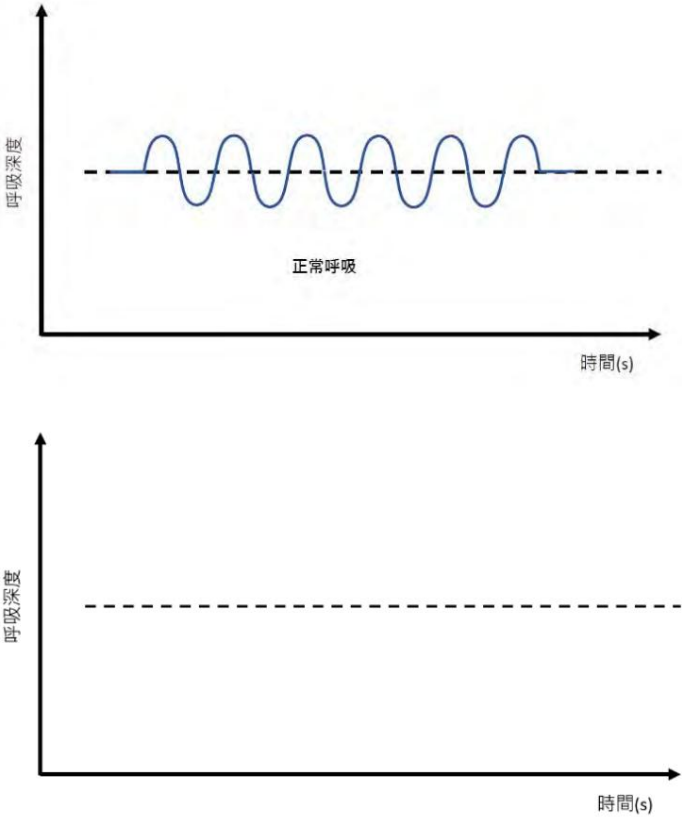
When the body's acid-base balance is disrupted, it can trigger a series of symptoms, such as metabolic acidosis, metabolic disorders, etc.

Alkalosis, respiratory acidosis, and respiratory alkalosis. Please answer the following questions.

1. Metabolic acidosis may be complicated by Kussmaul's breathing, as shown in the image below.

Please draw a diagram illustrating the normal respiratory rate and depth of Koosmore's breathing technique, based on the following figure.

Please provide a schematic diagram (2 points) and explain the mechanism (5 points).



2. Mechanism Explanation:

3. Which of the following factors can cause metabolic acidosis? Please select the correct option (2 points) and explain why.

(3 points)

- (A) Severe vomiting
- (B) Severe diarrhea
- (C) Hyperventilation
- (D) Insufficient ventilation
- (E) Traveling in the mountains

Explanation of the reasons for the options	

4. Which of the following factors can cause metabolic alkalosis? Please select the correct option (2 points) and explain why.

(3 points)

- (A) Severe vomiting
- (B) Severe diarrhea
- (C) Hyperventilation
- (D) Insufficient ventilation
- (E) Strenuous exercise

Explanation of the reasons for the options	

5. A-Ding and A-Hua went hiking on Nanhu Mountain (3,742 meters above sea level). Before they left, A-Cui reminded them to be careful of altitude sickness, which is caused by the body's inability to adapt to the special environment at high altitudes. Generally speaking, symptoms of altitude sickness will appear when climbing to an altitude of over 2,500 meters. As the altitude increases, the oxygen partial pressure decreases, and the incidence of altitude sickness also increases. Based on your understanding, which of the following symptoms will a person with altitude sickness experience (1 point), and why (1 point)?

(A) Metabolic acidosis (B)

Metabolic alkalosis (C)

Respiratory acidosis (D)

Respiratory alkalosis

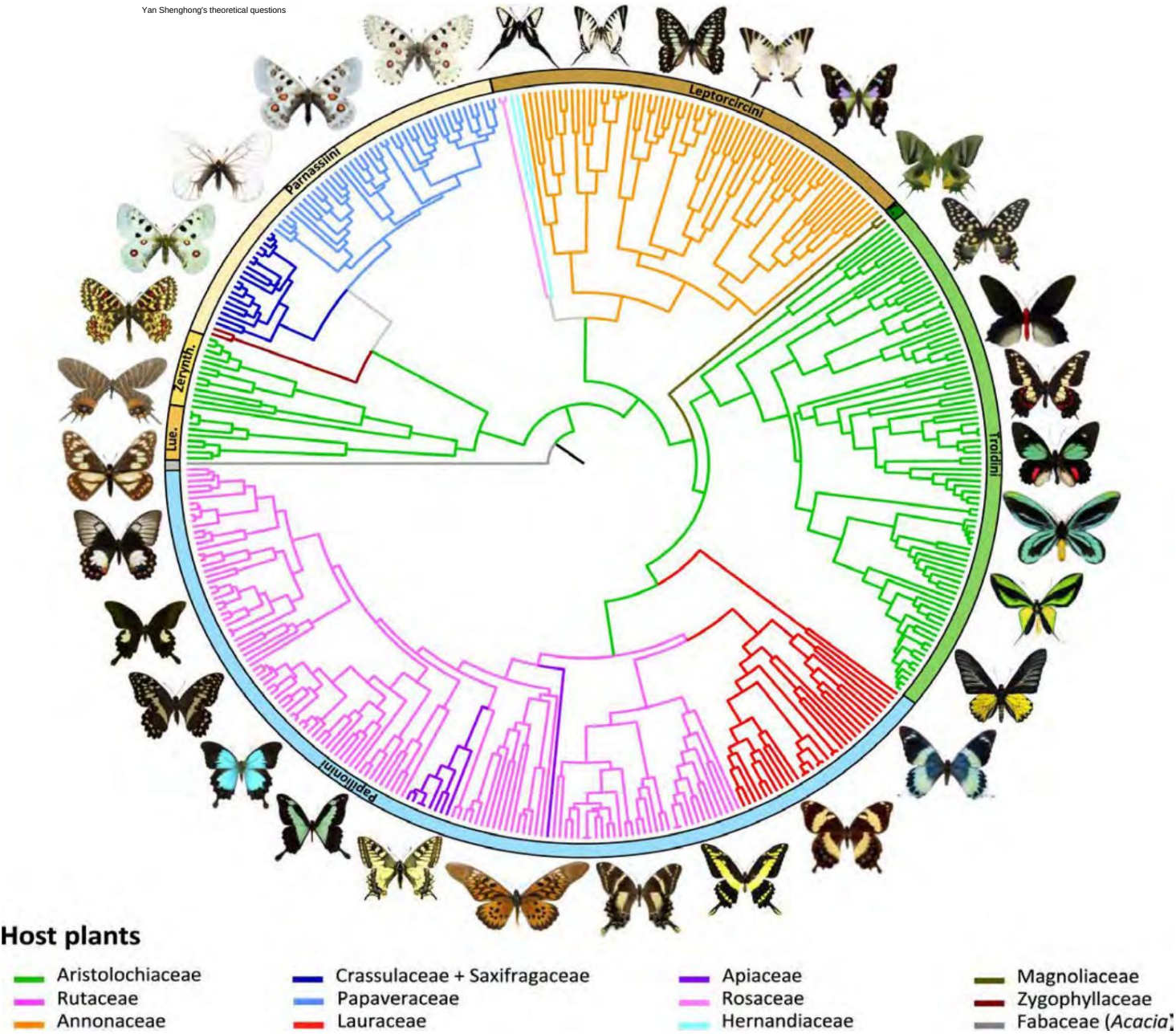
Explanation of the reasons for the options	

6. Following Acui's advice, Ahua prepared medication to prevent altitude sickness: Diamox.

Acetazolamide (CA) is a carbonic anhydrase (CA) inhibitor and diuretic. Why is acetazolamide used to treat altitude sickness? (4 points) Possible side effects include hyperventilation or hypoventilation (1 point). Why? (1 point)

IV. Multiple Choice Questions:

Yan Shenghong's theoretical questions



The image above is an evolutionary tree reconstructed from whole genome information for some species of the Papilionidae family worldwide. Each tribe is marked with a color, and the larval host plant preferences of each clade are also represented by different colors. Please answer the following questions based on this evolutionary hypothesis (25 points): 1. Which of the following statements is correct? (5%)

- (A) Sister groups of Troidini feed on Magnoliaceae. (B) Swallowtails utilize Aristolochiaceae four times in the evolutionary history of the Swallowtail family.
- (C) This diagram suggests that the Papilionaceae family has been the most successful in evolution among Rutaceae plants.
- (D) Brightly colored species feed on Aristolochiaceae, so Aristolochiaceae can provide a source of pigment synthesis, especially green and blue. (E) "Utilization of Apiaceae" appears twice in the evolutionary history of swallowtail butterflies, indicating that the second successful colonization was based on the experience of the first.

2. Among all plants, Aristolochiaceae, Crassulaceae, Saxifragaceae,

Secondary metabolites of Papaveraceae are thought to provide butterflies with toxic defenses against predators. Which of the following hypotheses is reasonable? (5%)

- (A) Leptocircini's flight speed is much higher than Troidini's. (B) Troidini's size is much larger than Parnassiini's because it feeds on Aristolochiaceae. (C) Although Parnassiini feeds on Crassulaceae and Papaveraceae, its white base color indicates that it is not... Warning coloration
- (D) The abundance of Leptocircini species in the environment must be less than that of Troidini for them to be protected.
- (E) Based on this hypothesis, it can be inferred that Fabaceae is the most primitive group among all the plants included in the discussion.

3. It is known that Luedorfini (abbreviated as Lue) is distributed in East Asia and Northeast Asia; Zeynthini (abbreviated as Zeynth) is distributed from the river valleys at an altitude of about 2000 meters on the Tibetan Plateau to Southern Europe; Parnassiini is distributed in the Holarctic region (North America, the northern half of Eurasia, and the Japanese archipelago); Leptocircini is distributed in Southeast Asia and Africa; and Troidini is distributed in North America, Central America, South America, Madagascar, East Asia, Southeast Asia, and Oceania (including Australia). Papilionini is distributed across all continents of the world. Which of the following hypotheses is reasonable? (10%)

- (A) Swallowtail butterflies were adapted to very cold climates when they originated. (B) The distribution of Zeynthini on the Tibetan Plateau suggests that their differentiation may have been influenced by the uplift of the Tibetan Plateau's strata. Influence
- (C) If Parnassiini is distributed throughout the entire northern region, it means that their migration may have taken place via the Bering Strait land bridge.
- (D) The Papilionini members that feed on Lauraceae were distributed in Central America, North America, and East Asia, which means that East Asia and the continent of Laurasia were once connected.
- (E) Troidini is distributed in Madagascar, Australia, and South America, so its differentiation must involve Gondwana. The split of the continent of Gondwana

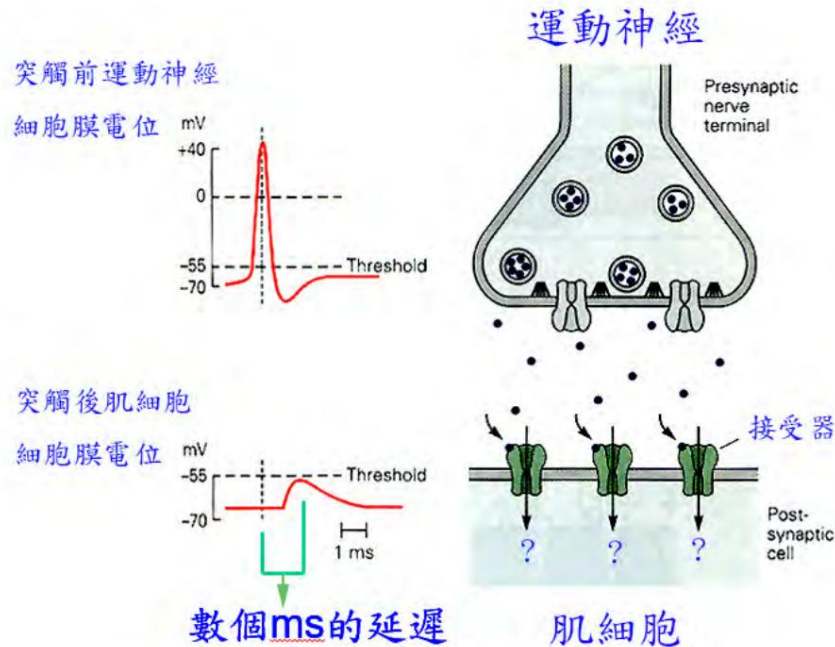
4. The information provided in the description in (3) assumes that the species sample size in this study has covered more than 95%.

Given the species of the Swallowtail family, which of the following inferences is reasonable? (5%)

- (A) Climate warming is a driving force for increased species diversity in Papilionidae. (B) Host plant shift is a driving force for increased species diversity in Papilionidae. (C) Host plant toxicity is a driving force for increased species diversity in Papilionidae. (D) Changes in flight ability and wing morphology are driving forces for increased species diversity in Papilionidae. (E) Changes in body size are driving forces for increased species diversity in Papilionidae.

V. Questions and Answers

The diagram below illustrates the synapse between a motor nerve and a muscle cell. After a delay of several milliseconds (ms) when the action potential reaches the nerve ending, the postsynaptic muscle cell records a change in the cell membrane potential indicating excitability. Please answer the following questions based on this information.



1. Why is the cell membrane potential (resting membrane voltage) of a nerve negative (-70 mV in the figure) when unstimulated? (5%)
2. Why does the action potential of a nerve cell rise to a large value instead of remaining at 0 mV during depolarization? (5%)
3. The depolarization of the action potential in nerve cells is caused by an increase in cell membrane voltage exceeding that of voltage-sensitive sodium ions. Why does the channel open to allow sodium ions to flow in at a threshold (approximately -55 mV in the figure)? Why does the polarization process pull the cell membrane voltage (which rises to nearly +40 mV at this point) back to the resting membrane voltage? (5%)
4. After the action potential reaches the motor nerve ending, there is a delay of several milliseconds before the excitatory postsynaptic membrane voltage change is recorded in the muscle cell. What are the reasons for this delay? (10%)

2022 Republic of China Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part Four Student Code



Precautions:

1. This section of the test consists of 4 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Four: (100 points)

1. During the cell division cycle, each chromosome first replicates into two sister chromatids in the S phase, and then in the M phase...

During this period, each pair of sister chromatids separates into different daughter cell nuclei. What is the mechanism by which the cell...

What mechanism ensures that each pair of sister chromatids separates correctly from each other? (25%)

2. Meiosis has two phases. During the first phase of meiosis, a pair of homologous chromosomes separate into different regions.

In the daughter cells, what is the mechanism that ensures a pair of homologous chromosomes separate correctly from each other? (25%)

3. In one animal, sex is determined by the sex chromosomes Z and W; females have ZW sex chromosomes, while males have ZZ.

WW cells cannot survive. Unfertilized eggs can sometimes develop into diploid offspring through parthenogenesis. Please...

Based on the developmental patterns of the following eggs, infer the sex ratio of parthenogenetic offspring.

A. The oocyte did not undergo meiosis, directly producing a diploid egg. (5%)

Male : Female = _____ : _____

B. Oocytes undergo only the first meiotic division to produce diploid eggs. (5%)

Male : Female = _____ : _____

C. After the oocyte completes meiosis, the chromosomes replicate once more to produce a diploid egg. (5%)

Male : Female = _____ : _____

D. After the oocyte completes meiosis, it produces an egg, which then randomly fuses with one of the three polar bodies to become a diploid.

(10%)

Male : Female = _____ : _____

4. A certain plant has two flower colors, white and red, regulated by two non-linked genes, A and B: Gene A has two...

Gene A has two alleles: dominant A and recessive a ; gene B also has two alleles: dominant B and recessive b. Dominant A

Dominant allele A and dominant allele B can express protein A and protein B, respectively; recessive alleles a and b are null alleles.

(null allele). Red flowers are due to the cells producing the red pigment RED; flowers that cannot produce RED are white.

RED can be synthesized via one of two different pathways: 1. The colorless precursor W1 is synthesized from protein A and protein B.

1. The combined action of protein B produces RED; 2. The colorless precursor W2 is produced through the action of other proteins.

RED, but protein A or protein B can inhibit this synthesis process individually. There are currently pure strains with two white flowers.

The F1 offspring of the hybrid all have red flowers. What is the reasoning behind this?

A. What are the genotypes of the parents of this double-flowered, purebred variety? (10%)

B. What is the genotype of the F1 generation of this hybrid? (5%)

C. If the F1 generation is self-crossed, what will be the ratio of red to white flowers in the F2 offspring? (10%)

Red flowers : White flowers = _____ : _____

Theoretical Assessment Test Questions for the **2022** Republic of

China Biology Olympiad

National Team Selection Camp

Part Five

Student Code



Precautions:

1. This section of the test consists of 9 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

I. *Batrachochytrium dendrobatidis*, a deadly fungus, has caused the deaths of more than 500 species of urticaria worldwide.

Habitat populations are declining, including the southern Darwin's frog (*Rhinoderma darwinii*) found in South America, whose survival rate is also decreasing.

The southern Darwin's frog is negatively affected by the chytrid fungus. Male southern Darwin's frogs possess a specialized vocal sac, which they use to carry the pathogen...

The sperm and egg are carried until the tadpole develops into an adult frog, at which point the tadpole is expelled from the adult frog's mouth and returned to the wild. This is because the tadpole can hear through its vocal sac.

This allows researchers to determine how many tadpoles each male frog carries, thus enabling them to obtain information without harming the frogs.

Knowing the number of offspring produced by each male frog, and combining this with the capture-and-release method, it is also possible to track the same male frog in different locations.

The study investigated monthly and annual variations in offspring numbers and estimated population fluctuations. Researchers conducted research in the South Pole from 2014 to 2019.

During the breeding season of the Southern Darwin frog (January and February each year), three sampling sites were selected in Chile for intensive research on the Southern Darwin frog.

The population survey included three sampling sites: one with a high rate of chytrid fungus infection, one with a low rate of chytrid fungus infection, and one with no chytrid fungus infection.

The sampling site was chosen to investigate the impact of **Tachytrium oryzae** infection levels on this frog population. (Adapted from Valenzuela-Sánchez et al.)

2022, Journal of Animal Ecology)

1. Researchers found that the average number of offspring per male frog in these three sampling sites was 4.5, 5, and 8.5, respectively.

Based on this data, researchers concluded that in sampling sites with high rates of *Dictyopharynx* infection, male frogs had the highest average number of offspring.

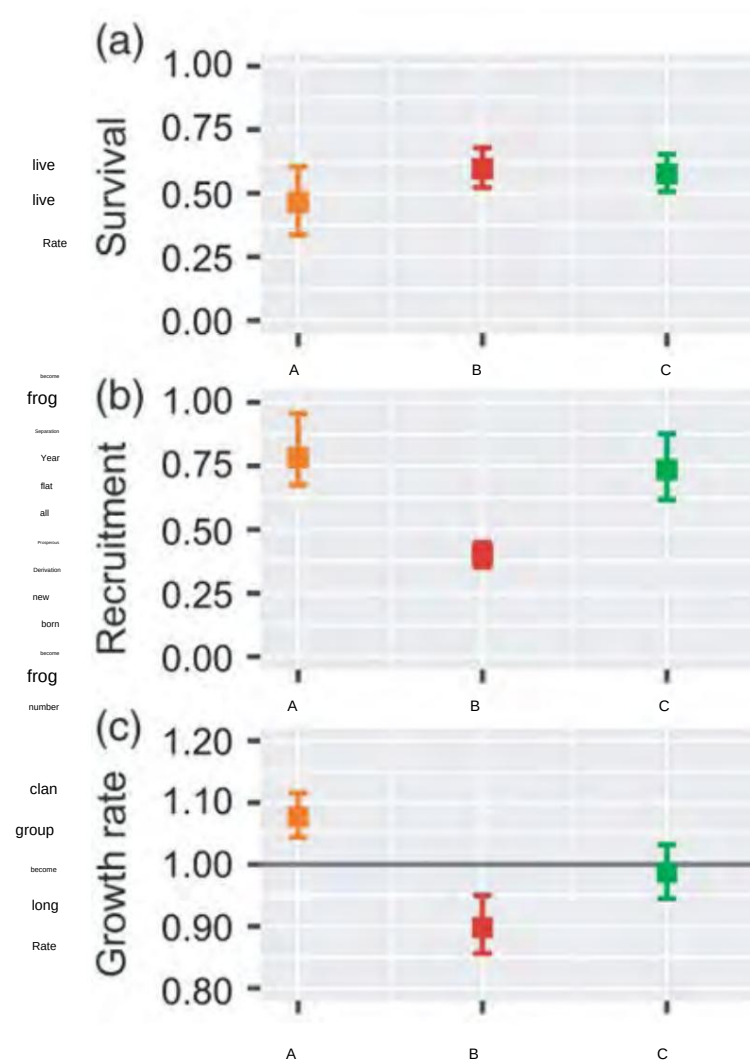
Sampling sites with low chytrid fungus infection rates also had the lowest average number of offspring in male frogs. Combining these two pieces of information, the prevalence of chytrid fungus infection in frogs was...

The results of the study on the degree and average number of offspring per male frog are presented in a bar chart. (8 points)

2. Following the previous question, based on the collected data, the researchers obtained the following additional results, where (a) represents the survival rate of adult frogs (assuming...).

(Assuming only the influence of chytrid fungi is considered), (b) is the average number of new frogs that each adult frog can reproduce in the following year, and (c) is...

Annual population growth rate. A, B, and C represent three sample points.



Based on the results above, answer correctly (O) or incorrectly (X) (True/False questions, 2 points each, 3 points deducted for incorrect answers).

(1) () Sampling site B is most likely the site with a high infection rate of *Cypridium*.

(2) () Sampling point A is most likely to be free of chytrid fungus infection.

(3) () Sampling sites with high infection rates of *Dictyopharynx* will see their populations grow year by year.

(4) () Without samples infected with chytrid fungus, the population size will decrease year by year.

(5) () The annual population growth rate is greatly affected by the survival rate of adult frogs.

(6) () The survival rate of adult frogs and the number of newly bred frogs in the following year are negatively correlated.

(7) () By increasing the number of offspring, adult frogs can maintain their population even if the survival rate is not high.

Group growth.

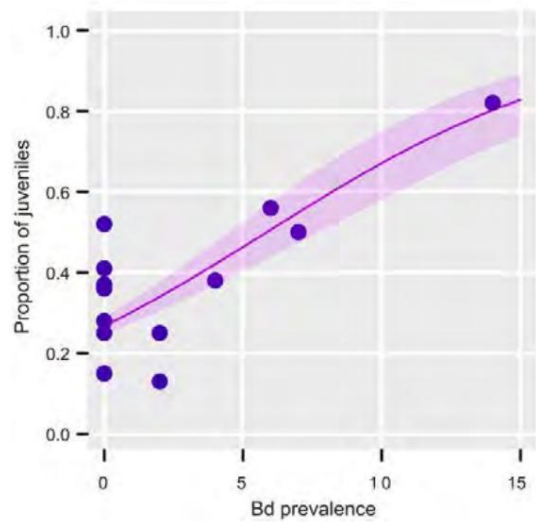
(8) Assuming that the sampling sites with high and low chytrid fungus infection rates have the same adult frog population density, then the two sampling sites...

The population growth rate of a point is mainly controlled by the population density (density-dependent).

(9) () Infection with chytrid fungus may cause adult frogs to allocate more resources to reproduction rather than individual growth.

3. Following on from the previous question, researchers investigated the infection status of *Dicrocephala glomerata* in southern Darwinian frogs at another 10 sampling sites (a total of 13 sampling sites). (Points), and divided adult frogs into two age groups, including subadults under one year old and adults over one year old, and obtained the results

The results are as follows. The horizontal axis represents the infection rate of *Dictyopharynx** at each sampling point, and the vertical axis represents the proportion of subadults to all adult frogs at each sampling point.

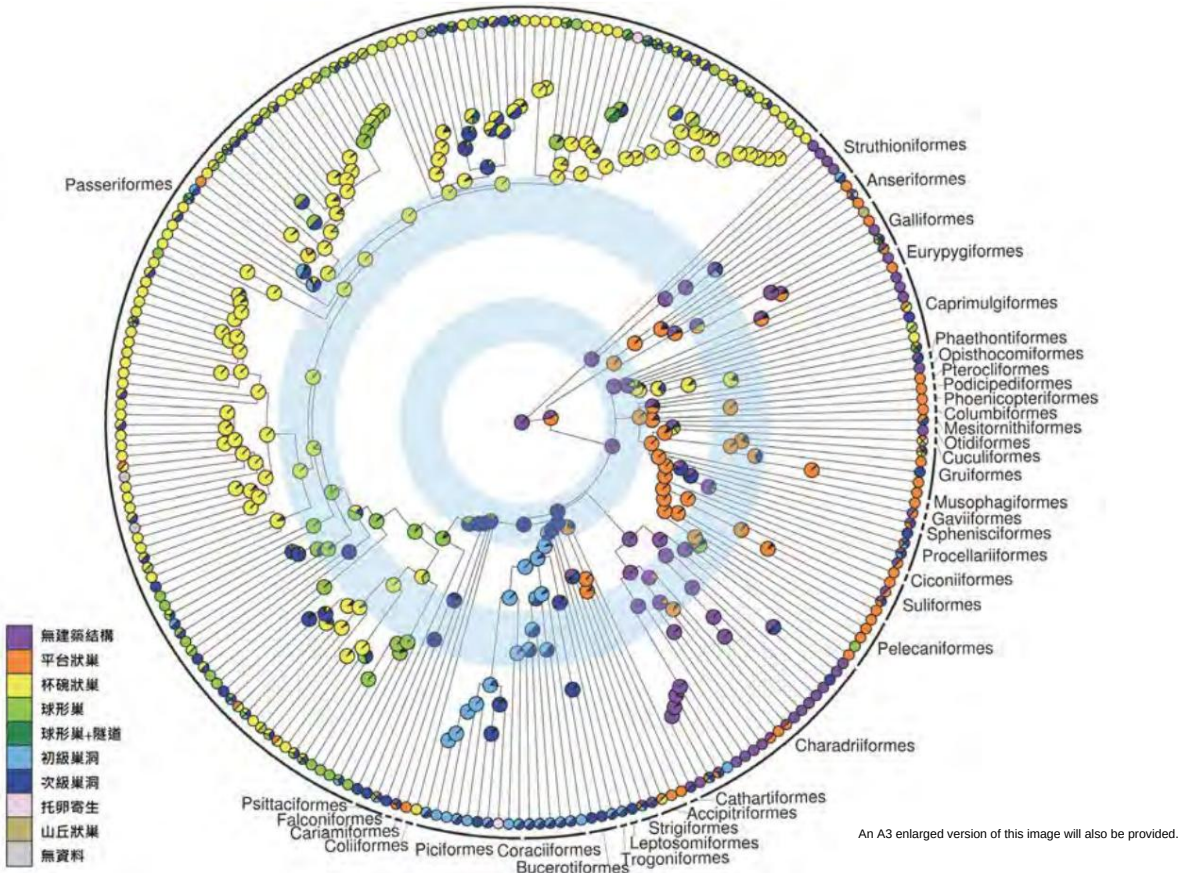


(1) Describe the relationship between the infection rate of *Trichytrium pratense** and the proportion of subadults in adult frogs in one sentence. (3 points)

(2) Infer the possible reasons for this phenomenon. (5 points)

II. The following diagram shows the relationship between the evolutionary phylogenetic tree of birds and the external structure of bird nests (adapted from Fang et al. 2018). This tree diagram...

Each endpoint represents one of 242 bird families, with the central endpoint representing the most recent common ancestor of all birds. Researchers (Fang et al. 2018) compiled data on the nest structures and nesting locations of over 5,000 bird species from 242 families worldwide, and then combined this with known bird species... A phylogenetic tree to explore the evolutionary history of bird nests. (Fang, YT, MN Tuanmu, & CM Hung. 2018. Asynchronous evolution of interdependent nest characters across the avian phylogeny. Nature Communication 9, 1863.)



1. These 242 bird families can be grouped into 36 bird orders. Which order should have the highest number of species? Please list them.

Write your name in the blank below. (Fill in the blank) (3 points)

2. Which of the following animals is most closely related to the Black-crowned Heron (belonging to the order Pelecaniformes)? (Single choice) (3 points)

- (A) Small Ringed Plover (belonging to the order Charadriiformes)
- (B) Crested goshawk (belonging to the order Accipitriformes)
- (C) Sparrow (belongs to the order Passeriformes)
- (D) Spotted dove (belongs to the order Columbiformes)
- (E) Little duck (belongs to the order Anseriformes)
- (F) Oriental White Stork (belonging to the order Ciconiiformes)
- (G) Cannot be determined

3. Which of the following animal orders first appeared on Earth? (Single choice) (3 points)

- (A) Small Ringed Plover (belonging to the order Charadriiformes)
- (B) Crested goshawk (belonging to the order Accipitriiformes)
- (C) Sparrow (belongs to the order Passeriformes)
- (D) Spotted dove (belongs to the order Columbiformes)
- (E) Little duck (belongs to the order Anseriformes)
- (F) Oriental White Stork (belonging to the order Ciconiiformes)
- (G) Cannot be determined

4. Which of the following animals was the last to complete its species evolution on Earth? (Single choice) (3 points)

- (A) Small Ringed Plover (belonging to the order Charadriiformes)
- (B) Crested goshawk (belonging to the order Accipitriiformes)
- (C) Sparrow (belongs to the order Passeriformes)
- (D) Spotted dove (belongs to the order Columbiformes)
- (E) Little duck (belongs to the order Anseriformes)
- (F) Oriental White Stork (belonging to the order Ciconiiformes)
- (G) Cannot be determined

5. Which of the following bird nest structures are the result of convergent evolution? (Multiple choice) (6 points)

- (A) Platform-shaped nest
- (B) Cup-shaped nest
- (C) Spherical Nest
- (D) Primary Nest
- (E) Secondary Nest
- (F) Oviparous parasitism

6. Based on paleontology, avian ecology, and this research, which of the following statements is correct? (Multiple choice) (10 points)

- (A) There are many birds with cup-shaped nests.
- (B) The closer the kinship between birds, the more similar the nests they build.
- (C) Birds that parasitize eggs have nests with different external structures.
- (D) Birds had already developed nests with architectural structures during the Mesozoic Era (251 million to 66 million years ago).
- (E) Nest diversity and bird habitat diversity are independent of each other.

7. Based on the diagram, please identify (A) a platform-shaped nest, (B) a spherical nest, (C) a primary nest hole, (D) a secondary nest hole, and (E) egg parasitism.

The order in which these bird nest structures appeared. Please fill in their codes in the blanks below in sequence (Fill in the blank) (5 points)

_____ Earlier _____ Earlier _____ Earlier _____ Earlier _____

III. The interaction between plants and soil microorganisms (generally referred to as plant-soil feedback, PSF) affects plant growth and survival, interactions between individual plants, and consequently the species composition of plant communities and soil microorganisms. PSF may change over time, thus affecting the dynamic changes of plants and microorganisms. To understand this, researchers analyzed the changes in the microbial composition of the root soil of four major dominant plant species in a Bodega Bay dune community in California as the plants aged. They also examined the impact of root soil microorganisms of different plant species on plant growth through greenhouse experiments.

The four plants in this study are *Ammophila arenaria* (Poaceae family) and *Carpobrotus edulis* (Aizoaceae family).

Baccharis pilularis (Asteraceae) and *Lupinus arboreus* (Fabaceae), the first two being introduced species and the latter two native to California. In a greenhouse experiment, researchers planted seedlings of these four plants in soil samples taken from the roots of these four plants, creating 16 combinations (4 plants $\times$ 4 soils). Each plant-soil combination included both unsterilized and heat-sterilized soil to understand the impact of soil microorganisms on plant growth. Furthermore, when collecting soil samples from the plant roots, researchers also selected soil samples from plants of different ages to understand the changes in root soil microorganisms as the plants grew and their impact on the plants.

Figure 1 shows the change in the number of fungal species in the root soil of these four plant species as the plants age. Figure 2 shows the change in the abundance of fungi of different functional groups in the soil as the plants age. Figure 3 shows the effect of the root soil of these four plant species on plant growth. Figure 4 further examines the effect of the root soil of plants of different ages on plant growth (excerpted from Ke et al.). (2021 New Phytologist). Please refer to Figure 1-4 for information and answer the following questions:

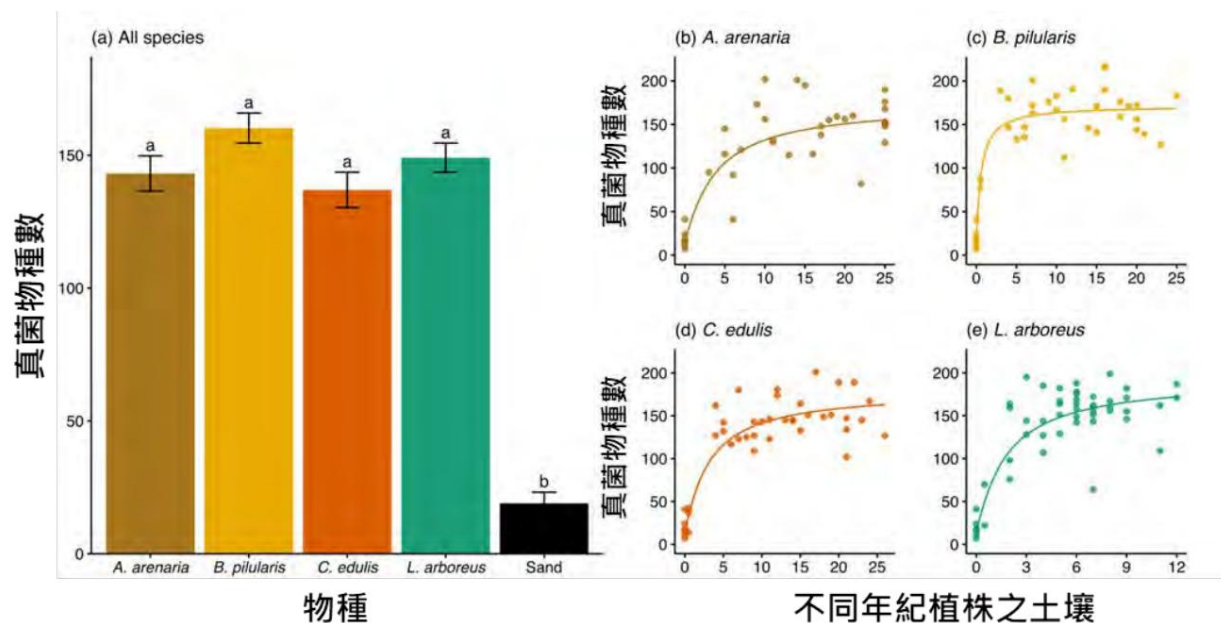


Figure 1. (a) Number of fungal species in the root soil of four plant species: *Ammophila arenaria**, *Baccharis pilularis**, *Carpobrotus edulis**, and *Lupinus arboreus**, where *sand* represents sand dune soil without plant growth; (b)-(e) show the changes in the number of fungal species in the root soil of these four plants as the plants age. The letters in Figure (a) represent the results of statistical tests; different letters indicate significant differences in the means.

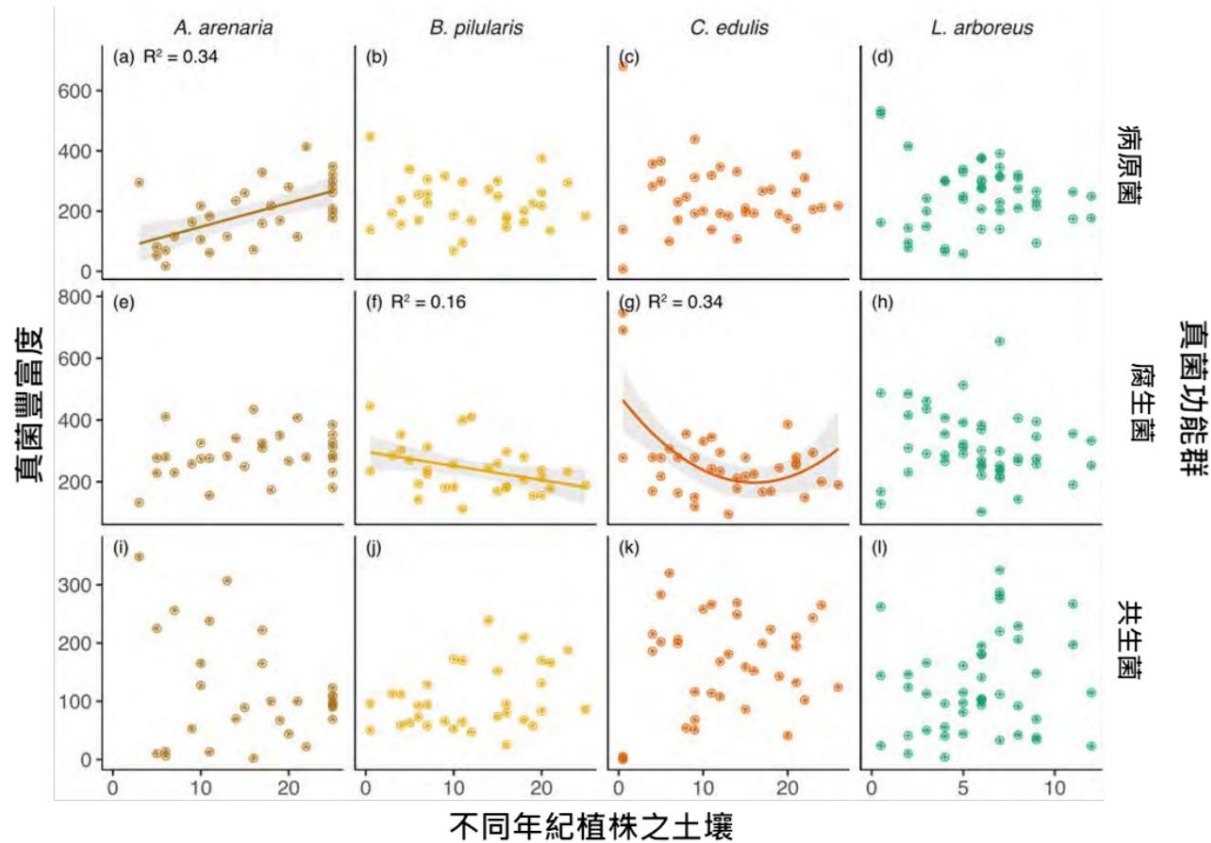


Figure 2. (Top) Changes in the abundance of different functional groups of fungi in plant root soil with increasing plant age. (a)-(d) are pathogens, (e)-(h) are saprophytes, and (i)-(l) are symbiotic fungi. Column 1 (a, e, i) is *Ammophila arenaria*, column 2 (b, f, j) is *Baccharis pilularis*, column 3 (c, g, k) is *Carpobrotus edulis*, and column 4 (d, h, l) is *Lupinus arboreus*. Trend lines indicate that the abundance of this fungal group changes significantly with increasing plant age. R^2 represents the explanatory power of the trend line for the data variation.

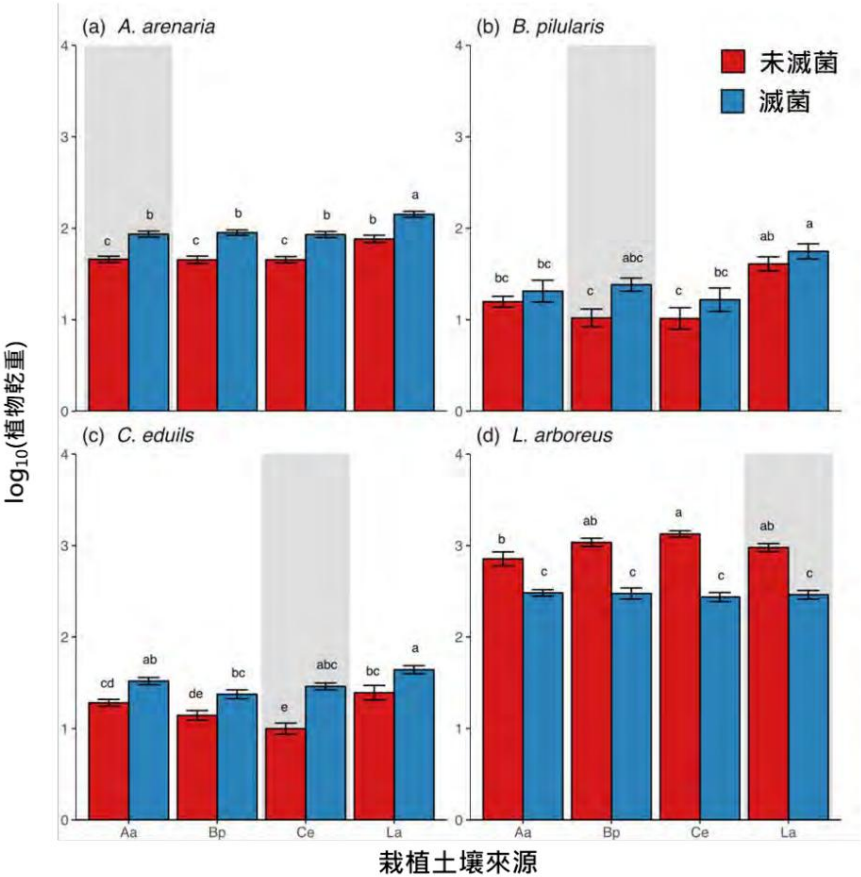


Figure 3. (Left) Average dry weight of individual plants planted in the root zone soil of different plant species. Plants (a)-(d) are *Ammophila arenaria* (Aa), *Baccharis pilularis* (Bp), *Carpobrotus edulis* (Ce), and *Lupinus arboreus* (La), planted in soils Aa, *Baccharis pilularis* (Bp), *Carpobrotus edulis* (Ce), and *Lupinus arboreus* (La), respectively. The bar chart shows root soil samples from Bp, Ce, and La plants (red bars) and root soil samples from the same plant species that have undergone high-temperature sterilization (blue bars). The error bars in the bar chart represent the standard error; the letters indicate the results of statistical tests, with different letters indicating significant differences in the means.

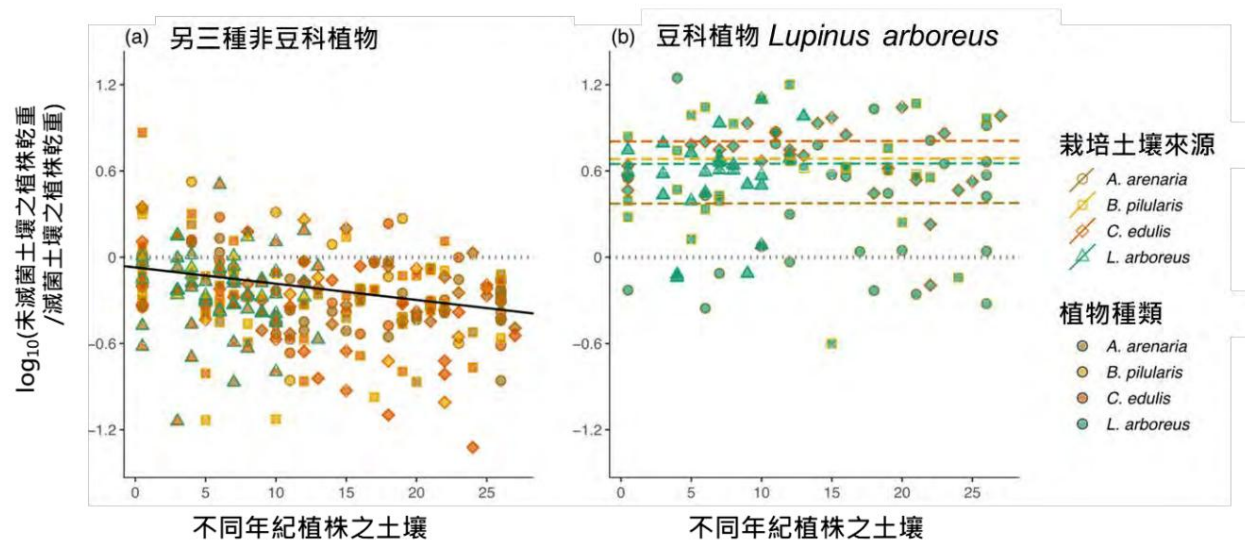


Figure 4. The effect of root soil on the growth of (a) non-leguminous plants and (b) leguminous plants (*Lupinus arboreus*).

The influence of this. The vertical axis in the figure represents the ratio of the dry weight of plants grown in unsterilized soil to that in sterilized soil, with different shaped symbols. The dots represent soil from different plant roots, with each dot's color representing a different plant species. The solid lines in the left image represent three different types of soil. The trend line of the combined analysis of experimental results for non-leguminous plants has a slope that is significantly different from 0; the long dashed line in the right figure represents L. The results of experiments on arboreus planted in the root soil of different plants showed that the slopes of all four trend lines were not significant.

True or False Questions: (33 points, 5 points deducted for each

incorrect answer) Which of the following statements is true? Please fill in • before the question number if it is true, and × if it is incorrect.

- ___ 1. As the plant ages, the number of fungal species in its root soil gradually increases, and the older the plant, the higher the number of fungal species.
The more fungal species are added within a given time period, the greater the increase in the number of fungal species.
- ___ 2. Among the four plant species, the soil around the roots of *Baccharis pilularis*, a plant in the Asteraceae family, had the highest number of fungal species.
It is taller than the other three types of plants.
- ___ 3. Most fungi in dune soils rely on resources provided by plants to grow.
- ___ 4. As the plant ages, the abundance of fungi in different functional groups gradually increases, along with the overall number of fungal species.
The trends of change are consistent.
- ___ 5. As the plant ages, the abundance of fungi that live in symbiosis with the legume *Lupinus arboreus* gradually increases.
And promote the growth performance of the plant.
- ___ 6. For the grass *Ammophila arenaria*, planting in the root zone soil (Aa) of the same species has the best effect.
Excellent growth performance.
- ___ 7. For the Aizoaceae plant *Carpobrotus edulis*, planting it in the root zone (Ce) of a plant of the same species has the best effect.
Poor growth performance.
- ___ 8. Sterilized soil, lacking the mycorrhizal fungi that coexist with plants, is not conducive to the growth of *Baccharis*, a plant in the Asteraceae family.
The growth of *pilularis*.
- ___ 9. The growth performance of non-leguminous plants planted in soil containing plants of different ages may be related to the abundance of pathogens in the soil.
It is related to the changing trend.
- ___ 10. For two introduced plant species in the California dune flora, as the plants age, the soil microorganisms suppress...
This inhibits their growth, thereby limiting the expansion of these two invasive plants.
- ___ 11. Sterilized soil is not conducive to the growth of the legume *Lupinus arboreus*, possibly because the soil lacks nutrients.
Rhizobia that help it fix nitrogen.